Fastwind v10/11

User Manual

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Introduction – Brief Overview

Fastwind is a NLTE, unified stellar atmosphere/spectrum synthesis code that allows to calculate the structure of hot massive star atmospheres incl. line-driven wind and their emergent spectra in a robust and fast way. The temperature stratification is calculated from the electron thermal balance, and there are many possibilities to include wind inhomogeneities (wind clumping) for clumps assumed to be optically thin(v10/v11) or of arbitrary optical thickness (v10 only), see `subroutine clumping` in `nlte.f90`. Moreover, the X-ray emission from wind embedded shocks can be simulated and included into the calculation of the ionization equilibrium. Various possibilities to calculate the shock structure (in particular, the temperature run) can be considered.

At current time (2026), the available program package allows to set up either Fastwind v10.8 (very fast, due to specific approximations, for details see Santolaya-Rey et al 1997, Puls et al. 2005, Rivero González, et al. 2012) or Fastwind v11.8 (still fast, but slower than v10.8, because of less approximations, see Puls et al. 2020) by means of the same programs. Which of both versions will be generated depends exclusively on the state of the `parameter optcmf_full(module nlte_opt` in `nlte.f90)`, which, if set to `T,` results in `v11`, and, if set to `F`, in v10.

The main philosophy of Fastwind is tied to two approaches:

(i) whenever possible, use fast numerical tools or methods, even if slightly approximate.

(ii) split the elements in two groups, the so-called explicit ones (or foreground elements), and the background elements, where the former group should consist of those elements which are specifically important for the spectroscopic analysis aimed at (e.g., H, He, N for hot stars or H, He, Si for B-type stars). Treat this group (both with respect to atomic data and the setup/solution of the rate equations) as precise as possible. Moreover, use a flexible atomic data input format (DETAIL, Butler und Giddings 1985, see enclosed material: "`detail.pdf`"). On the other hand, try to calculate the background elements (typically including Fe and Ni) with fast methods, and use a fixed atomic data base (from WM-basic, Pauldrach et al. 2001).

In our approach, most explicit and background-element levels are *packed*, see next section.

The major difference between v10 and v11 relates to the radiative transfer. When calculating the line rates, in v10 only lines from explicit elements plus the strongest lines from the background ones are treated in the comoving frame (CMF), whilst all other bound-bound transitions are treated in the Sobolev approximation. The radiation field itself, however, is quantified based on a pseudo-continuum approach, including approximate line opacities and emissivities. In v11, on the other hand, all lines in between the range `wavblue` to `wavred` (typically from 200 to 10000 Å) are treated in the CMF, and the other frequency regions as in v10.

Level packing

Let us describe already here one of the major features of our current approach: most of the atomic levels (both explicit and background elements) have been *packed*, to allow for fast turn-around times in the NLTE calculations, and for a numerically (though not physically) more precise solution[1], compared to the case when all levels would be treated individually. Contrasted to the *super-level* approach (Anderson 1990) as adopted, e.g., in the alternative codes CMFGEN (Hillier and Miller 1998) and PoWR (Gräfener et al. 2002), where **within** each super-level the occupation of the sub-levels is assumed to be in LTE,

$$\frac{n_j}{g_j} = \frac{n_i}{g_i} e^{-h\nu_{ij}/kT}$$

(with occupation number density n_i and statistical weight g_i , and other quantities with their usual meaning), in our current *packing* approach (most) sub-levels are chosen in such a way that they belong to the same multiplet **and** that their energy difference $h\nu_{ij}$ is low compared to kT. Then, for all individual sub-levels within the considered multiplet, the ratio n_j/g_j is (almost) the same, and identical to the ratios

$$\frac{n_j}{g_j} \cong \frac{n_i}{g_i} \cong \frac{n_{tot}}{g_{tot}},$$

where the latter is the n/g ratio for the packed levels, with $g_{tot} = \sum g_i$ and $n_{tot} = \sum n_i$ (prove the last equality in the equation above by yourself). Thus, in the NLTE framework, we solve for the packed levels [in Fastwind, for n_{tot} (when considering explicit elements) and for n_{tot}/g_{tot} (when considering background elements), respectively].

In the radiative transfer (to calculate the radiative line rates), we either consider one representative transition (v10) with a total oscillator strength derived from

$$\sum g_i f_i := g_{tot} f_{tot} \quad \text{such that} \quad f_{tot} = \frac{\sum g_i f_i}{\sum g_i},$$

or (in v11) we consider the individual components, using $n_i/g_i = n_{tot}/g_{tot}$, and the appropriate individual f_i-value. Then, the individual opacities are (without stimulated emission)

$$\chi_i \propto n_i f_i = \frac{n_i}{g_i} g_i f_i = \frac{n_{tot}}{g_{tot}} g_i f_i = n_{tot} \frac{g_i f_i}{g_{tot}},$$

i.e., they can be expressed in terms of the total (packed) occupation numbers, the individual gf-values, and the total (packed) statistical weight, cf. Chap. I/VII/5. How to pack the resulting mean intensities (from the radiative transfer of the individual components) such that they correctly enter the rate equations for the packed levels has been decribed in Puls (1987).

[1] for such packed, fewer levels, the linear rate equation system becomes numerically better conditioned

For future calculations, if certain levels (e.g., from Fe) should be calculated with a higher precision, we suggest to incorporate the super-level philosophy into Fastwind (see 'To Do List').

High-lying upper levels of background elements

Because of computational and numerical limitations, high-lying ionic levels are treated in an approximate way. In the NLTE approach, we assume that these levels are sparsely populated, and neglect them in the rate network. For the radiation field, however, transitions connected with such levels can be important, particularly, if the lower level is a ground level or a low-lying metastable one. Thus, for neither opacities nor source functions these transitions (with an index of `iup=0` for the upper level) can be neglected. The opacities are then calculated by discarding stimulated emission, and the source functions are approximated by a two-level atom, where the irradiation results from the (pseudo-) continuum radiation field (v10) or the smoothed CMF solution (v11), see Chap. I/III.

In **v11**, and if the logical parameter `opt_modify_sl_noup (module nlte_opt)` is set to `.True.`, which is the default, the (two-level) source function is allowed to become modified by overlapping transitions from explicit elements[2] (see top of `subr. overlap_noup` in `cmf_all.f90`). Here, the original two-level-atom approach can be restored when the latter parameter is set to .False. (recompile!).

[2] In the current version, only lines from non-HHe explicit elements such as N are considered. This might need to be investigated and modified in future versions.

A quick guideline of how to calculate a specific model

This section is intended as a quick guideline of how to calculate a specific atmospheric model/synthetic spectrum with Fastwind, where all details are covered in the text following this section. Here, we assume that the code has been installed in a file structure as described in Chap. I/I, and that we are working in one of the *working directories* as exemplified there.

- copy the intended solar abundance file to `abundan_solar` in `inicalc/ATOMDAT_NEW.`
- check whether the options/parameters in `nlte.f90` are set correctly (particularly, regarding `optcmf_full`, which controls whether v10 or v11 will be running, and `fpath1`, which controls which data for background elements are chosen).
- under very specific conditions, check whether the number of grid points and impact parameters (`nd, np`) needs to be enlarged in `dimes.f90`
- check whether `Makefile` points to the correct `LINES**.dat` file (close to end), otherwise modify. You might also need to modify the default atomic data file (A*.dat).
- compile the code package (`make clean`, and then `make`)
- copy the input files for atmospheric structure and formal integral to the local *working directory*, and check/modify them:
    - `INDAT.DAT` – in particular name of *model directory*, abundances, and clumping properties
    - `FORMAL_INPUT` → which line complexes and/or spectral ranges?
    - if new spectral lines to be included, update also `LINES**.dat in inicalc/DATA.`
- create an empty directory within the local *working directory*, with a name as in `INDAT.DAT` (1st line). This will be the *model directory*, where all output data will be stored.
- run
    - `pnlte_A*.eo`
    - `ptotout_A*.eo`
    - `pformalsol_A*.eo`

- check `FASTWIND.LOG` or specific files in the *model directory* whether all calculations have ended without error, and whether calculations have converged (see Sect. I/IV/3).

Part I. General set-up and calculations/results

The following issues refer to both v10 and v11.

Download the following directories:

`inicalc`: **must** be downloaded

`inicalc_ATOMDAT_NEW_old`: if comparison with previous versions aimed at.

`inicalc_ATOMDAT_2016_new`: if extended atomic structure for low ionization stages to be tested

`make_ATOMDAT_NEW_new`: if data base (bg elements) `ATOMDAT_NEW_new` (installed by default) shall be updated

`make_ATOMDAT_2016_new`: if data base (bg elem.) `ATOMDAT_2016_new` shall be updated

Chapter I/I. Recommended directory structure

main directory: `NLTE`

- should contain `inicalc` (where all the original stuff resides, copied from download location)

and, after copying from inside `inicalc`

- *in case*, `idlproc`: `IDL`[3] (or `GDL`[4]) procedures to plot atmospheric structure, occupation numbers, and SED, incl. `startup.pro` (initialization) or `startup_gdl.pro`, respectively.
- file `HOPFPARA_ALL_HHe` (see below)
- file `HOPFPARA_ALL_met` (see below)
- *in case*, subdirectory `inicalc_ATOMDAT_NEW_old`: atomic data for background elements, previous version (affected by few problems): see `make install_data` (Chap. I/III)

[3] Interactive data language
[4] Gnu data language (for free), similar to IDL; many IDL procedures can be run without problems

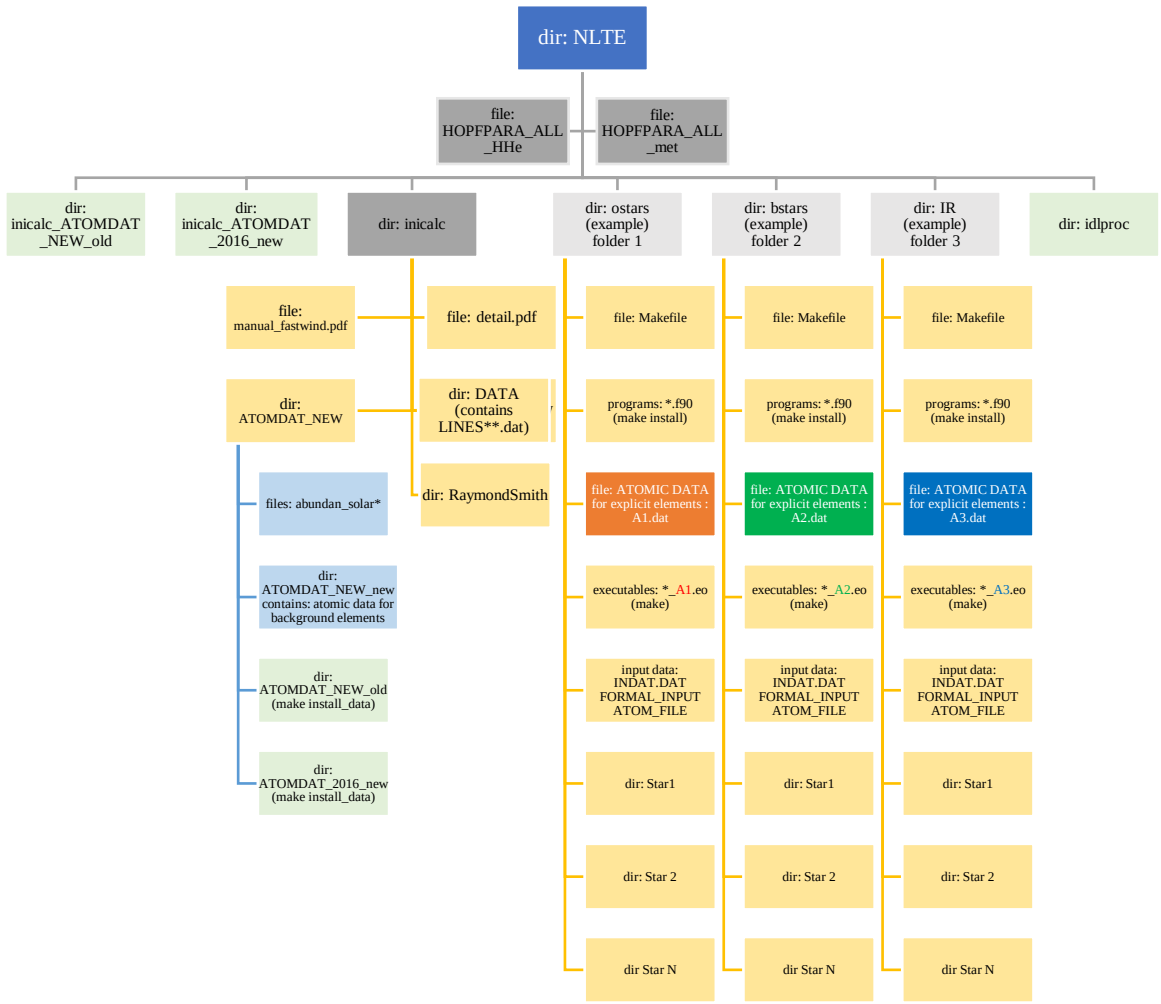

Figure 1: Recommended directory structure (see text). The directories "ostars" etc. will be denoted as *working directories*, and the directories "Star 1" to "Star N" as *model directories*. Directories in light-green are **not** mandatory.

- *in case*, subdirectory `inicalc_ATOMDAT_2016_new`: atomic data for background elements, with extended low ionization stages (not tested): see `make install_data` (Chap. I/III)

(sub)directory `inicalc` **contains** (among other files/directories not further mentioned)

- `manual_fastwind.pdf`: this manual
- `detail.pdf`: a comprehensive description of the numerical structure of the *explicit* atomic models
- `subdirectory ATOMDAT_NEW` (substantially changed compared to previous versions)

  atomic data for *background* elements (various versions)

- `ATOMDAT_NEW_new`: **default data base**, set by variable `fpath1` in `module fastwind_params (nlte.f90)`, see below, contains
  - `atomnl2_meta_v2.0` (for `nlte_approx v6.0` and higher)
  - `atomcol, atomgf3, generalinfo, ionisen, partit`
  - files for (background) metal-lines (Chap. I/II), typically denoted by `nl3*_all`
  - `atomnldr` (data for dielectronic recombination – DR, from v10 on)
  - and few more files, which are (currently) not required for the actual calculations
- `ATOMDAT_NEW_old`: data base for previous Fastwind versions (until v10.7.1/11.7.1), affected by few errors, see below.

  Not present after download, to be installed with `make install_data` (see Chap. I/II). Only required if comparison with previous results aimed at. Compared to the other two data bases, more files present (including: old version of `atomnl2_meta_v2.0`, line lists in ascii, README instructions and various programs). Details can be found below and particularly in Part IV of this manual.
- `ATOMDAT_2016_new`: **Not present after download,** to be installed with `make install_data` (see Chap. I/II). Content similar to data base `ATOMDAT_NEW_new` described above, but slightly different *and larger* – lower ionization stages improved compared to `ATOMDAT_NEW_new`. Still not decently tested (at least one error present, see below).

We recommend to use `ATOMDAT_NEW_new`, which is the default (`fpath1` in `module fastwind_params`) and already installed from the beginning on. This base is very similar to the original one delivered in previous Fastwind versions (still present in `ATOMDAT_NEW_old` if installed), but with improved F I, Cl I, N VI, and Al III.

Note that data in `ATOMDAT_2016_new` have been particularly updated with respect to lower ions (stage I, II), but certain deficiencies need to be improved in future versions. Most obvious are wrong wavelengths for the S IV UV-resonance doublet at approx. 1063/1073 Å, which appear at 1067/1078 Å in this base. This problem is not present in directory `ATOMDAT_NEW_new` (lower number of lines from low ions). Note as well that the data in `ATOMDAT_2016_new` are, except for improved Cl I and Al III, consistent with data displayed in

https://www.usm.uni-muenchen.de/~cvs/Atomic-Models-Source/html/index.html.

Note that most current and potentially updated data can be also visualized (after running `script_nl3`, see below) via the html-file `make_ATOMDAT_*/fw_data/html/index.html`, where * is either `New_new` or `2016_new`, and `make_Atomdat_*` are two separate packages delivered together with `inicalc`, required for updating the atomic data base for background elements (see Part IV).

subdirectory `ATOMDAT_NEW` also **contains**

- `k_shell_Auger_data` (for Auger ionization and K-shell absorption)
- `abundan_solar` (which is actually used); modify or copy your preferred solar abundances to this file, e.g., from ….
  - `abundan_solar_old` ("old" values, from Grevesse and Sauval 1998)
  - `abundan_solar_asplund_2009` (recommended, following Asplund, et al. 2009)
  - `abundan_solar_asplund_2021` (recommended, following Asplund et al. 2021)
  - `abundan_solar_mean_light_ion` (for tests with mean light ion following Mihalas 1972)

  To be on the safe side, check, after running the code for a very short time (less than one second), the `ABUNDAN` file in the *model directory*, whether everything is as intended.

All these data are required if background metal-opacities shall be calculated.

(sub)directory `inicalc` contains as well

- subdirectory `DATA`
  - `LINES**.dat:` metal line broadening data for calculating the formal integral, and data for depacking specific transitions when calculating the line opacities in v11. Examples for **
    - ** = Si; data for silicon (rather old model atom)
    - ** = N2345; data for nitrogen (from Rivero Gonzalez et al. 2012)

        - ** = CnewNOSi_new_coll; data for carbon, nitrogen, oxygen, silicon (currently recommended)
        - and many more variants
    - `thom_new.dat` line broadening data for H/He (following Schöning and Butler 1989)
    - U functions (also for level inversions): coupling to continuum in Sobolev calculations (see Hummer and Rybicki 1985 and Taresch et al. 1997, Appendix A).
        - `U2B.dat`
        - `U2L1.dat`
        - `U2L2.dat`
        - `U2S.dat`
        - `utablag.dat`
        - `utablak.dat`

(sub)directory `inicalc` contains as well

- `OP_DATA_NEW`: ionization cross sections from Opacity Project (OP)
- `RaymondSmith`: data for X-ray emission calculated in `lambda_xray.f90`
- subdirectory `newprog`: current and
- subdirectories `newprog_vx.x:` previous program versions

Working directories (within `NLTE`, **in parallel to** `inicalc`):

`astars, ostars` etc., **to be created by the user**. Already here, we point out that these different subdirectories are required when using different atomic data files (`*.dat`, see below) which result, after compilation, in different executable objects. E.g., a program compiled with `A1.dat` will result in an executable object `pnlte_A1.eo`, `pnlte_approx_A1.eo` etc., whilst another directory with atomic data `A2.dat` will contain, after compilation, the different files `pnlte_A2.eo, pnlte_approx_A2.eo` (if A2 were different from A1). The reason for working with different model atoms is the need that a spectroscopic analysis of different spectral

types usually requires the analysis of different elements/ions (e.g., Si instead of He/N for B-types vs. O-types).

Anyhow, these directories will contain, **after preparing/copying** …

… all program (see below) and data files that are needed to calculate models, occupation numbers, fluxes and profiles, in particular the `Makefile` (intel=ifort, gfortran).

`Makefile` needs to be copied from `Makefile_v11_gfortran` or `Make-file_v11_ifort` in `inicalc`, and is also valid for v10.

install all **programs** (*.f90) with **`make install`** or by simply copying (all from "`inicalc/newprog`")

- `nlte_type.f90, dimes.f90, ffr.f90, princesa.f90`
- `nlte.f90, nlte_approx.f90, lambda_xray.f90, cmf_all.f90`
- `totout.f90`
- `preformal.f90, formalsol.f90, formalsol_add.f90`

in case, install the additional data bases for the background elements (`ATOMDAT_NEW_old` and `ATOMDAT_2016_new`) into `inicalc/ATOMDAT_NEW`, with **`make install_data`**. To this end, the directories `incalc_ATOMDAT_NEW_old` and `incalc_ATOMDAT_2016_new` have to be present (download!), see Fig. 1.

… one of the following data files (copy again from `inicalc`!)

**.dat: atomic data for explicit elements

** = `A10HHe, A11HHe, A10HHeSi, A10HHeN2345, A10HHeCnewNOSi`

and the user supplied input files (copy and modify for your purposes, described in detail later on)

`INDAT.DAT` (for `nlte`)

`FORMAL_INPUT` (for formal integral), copied from
`FORMAL_INPUT_A10HHe` or
`FORMAL_INPUT_A10HHeSi` or
`FORMAL_INPUT_A10HHeN2345` etc.

Every new model is calculated in a subdirectory **below** the current *working* directory (`astars, ostars` etc.), and all the calculated data are stored there. In the following, this subdirectory will be called *model directory*, and needs to be created (empty) by the user.

For logistic reasons, it is reasonable to have *one working directory* (e.g., `astars, ostars` etc.) per atomic model file. E.g., for hot O-stars, one might use `A10HHeN2345.dat` within `folder1`, for IR-work one might use `A11HHe.dat` within `folder2`, and for B-stars one might use `A10_HHeSi.dat` within `folder3`, where `folder1` to `folder3` are directories in parallel to `inicalc` (of course, the folder names should be chosen in a sensible way appropriate for the user's purposes). In our example, all three directories might contain hundreds or more *model directories* for massive stars with different parameters.

Note that when the calculations have been finished, specific files, in particular the file `METAL_RESTART`, can be deleted from the *model directory* if no restart is needed (this file is particularly large), as well as files `OPA_EFF_RAT_LA` and `xj_cmf_all` (the latter only for v11). For further files that can be deleted, see Chap. II/5 and III/4.

Chapter I/II. Generation of binary line-files

In order to create the required binary line files (in case of a non-Linux OS such as MS Windows), proceed as follows:

1. for the original data delivered in **previous** Fastwind versions (until v10.7/11.7) as still present in `inicalc/ATOMDAT_NEW/ATOMDAT_NEW_old` (when installed; otherwise in directory `inicalc_ATOMDAT_NEW_old`, requires download), follow the instructions in `README_conv_ascii`

2. For the almost identical (but updated regarding few ions, see Chap. I/I and Part IV) data set contained in `inicalc/ATOMDAT_NEW/ATOMDAT_NEW_new` (already installed in the download version), the required ascii data (`nl3*_all_ascii`) can be found within the separate package `make_ATOMDAT_NEW_new`, in directory `dir_atomnl2_meta`, **after `script_nl3` has been executed** (see Part IV of this manual).

3. `script_nl3` also generates new binary files from the above ascii files, via program `conv_ascii_add_HHelines.f90` (the `*.eo` version has been compiled for Linux and needs to be recompiled on a non-Linux OS), which are saved in directory `fw_data_new` under filenames `nl3*_all` (see below).

4. For the somewhat newer (but almost untested) dataset `ATOMDAT_2016_new`, the same procedure(s) can be applied, though within the package `make_ATOMDAT_2016_new`.

If one of these data bases will have been updated (because of newer or additional atomic data) , the (modified) ascii files will be automatically re-generated in `dir_atomnl2_meta`. Indeed, **all** background data files required for Fastwind will be created with `script_nl3` (see Part IV), and saved in `fw_data_new`. Of course, the script and the programs need to be adapted in case of a non-Linux OS (see top of `script_nl3)`. Don't forget to copy all updated files to the appropriate directories in `inicalc/ATOMDAT_NEW/ATOMDAT_*`, with `* = New_new` or `New_old` or `2016_new`.

Anyhow, in most cases (when using Linux), the above conversion is not required, and you can directly use the binary files, as provided in

`inicalc/ATOMDAT_NEW/ATOMDAT_NEW_new` (installed by default),
`inicalc/ATOMDAT_NEW/ATOMDAT_NEW_old` (if installed),
`inicalc/ATOMDAT_NEW/ATOMDAT_2016_new` (if installed),

in dependence of the path variable `fpath1` (see above):

- `nl3a_all` (wavelength)
- `nl3g_all` (gf values)
- `nl3i_all` (indices denoting lower and upper levels, e.g. a transition of CIV, level 1 to 2 has the index 6400102)
- `nl3info_all` (file-structure)

Chapter I/III. Generation of executable objects (= compilation and linking)

In the following, we assume that you have copied all programs (either handish or by "make install", see above) into the appropriate *working directory*.

Specific issues

- *for specialists:* If you want to calculate without approximate Helium, uncomment the line `jatom(2)=0` in `nlte_approx.f90, subr. atomdat` (this has no influence on the exact treatment, and is required only in those cases when there should be no Helium at all, i.e. neither in the exact case).
- If you want to allow for outer electron temperatures below 0.4 T_{eff}, change the parameter `TMIN1` in `nlte.f90, MODULE fastwind_params`

- If you want to use different stratifications of clumping factors, modify `subroutine CLUMPING` in `nlte.f90`

- **Always** (see above): in `inicalc/ATOMDAT_NEW`, copy one of the suitable solar-abundance files, `abundan_solar_*`, to the file used by the program, `abundan_solar` (in the same directory). If you want to change these abundances additionaly, change `abundan_solar.`

- If you want to calculate dwarf atmospheres with larger gravities, it might be useful to use more photospheric grid points than by default. Change the corresponding paramenters `(nd, np)` in `dimes.90`

Background elements: Radiative transfer, NLTE, and selected species

Both the radiative transfer and the NLTE treatment of the background elements varies from case to case:

- **v10 –** NLTE: for so-called non-selected elements, approximate NLTE (Puls et al. 2005); for so-called selected elements, exact NLTE, i.e., solution of rate equations. 'Selected' elements are defined in `subr. select (nlte_approx.f90)`, and comprise, by default, the elements C, N, O, Ne, Mg, Si, P, S, Ar, Fe, Ni, with *locally* three or four ionization stages in parallel.

- **v10 –** radiative transfer, to calculate radiative line rates:

  - simplified Sobolev transfer for levels with approx. NLTE (non-selected)

  - Sobolev transfer (incl. continuum) for transitions between levels of selected elements that are not too strong

  - plane-parallel static or spherical CMF transfer for strong lines (roughly a couple hundred) from selected elements

- **v10 –** radiative transfer, to calculate pseudo-continum fluxes (line-blanketing/blocking), photospheric radiative acceleration, and incident radiation for line transfer:

  - statistical treatment of lines, following Puls et al. 2005

- **v11 –** NLTE: as v10, except for approximate NLTE that uses, as incident radiation, the CMF-radiation field (smoothed)

- **v11 –** radiative transfer to calculate radiative line rates, emergent fluxes, radiative acceleration, etc.

- CMF transfer in the range `wavblue … wavred` with total opacities and emissivities from selected **and** non-selected (and explicit) elements. Outside this range, as v10.

NOTE: if additional elements (e.g., Al or Ca) were requested with higher accuracy than possible for non-selected elements, include them in the corresponding list as to be selected (`subr. select`). No further action required.

1. If you want to change the location of data files …

"Standard" users:

Leave them where they are, i.e., in `inicalc/DATA`, `inicalc/ATOMDAT_NEW`, etc. (see below).

If you really want to change the locations:

a. edit the file `nlte.f90, Module fastwind_params`: change the parameters `fpath_ufunc, fpath, fpath1, oppath, fpath_xrays` to the actual location (either relative or absolut) where you have saved the directory `DATA (fpath_ufunc), ATOMDAT_NEW (fpath), ATOMDAT_NEW/ATOMDAT_NEW_new` (or similar, `fpath1), OP_DATA_NEW (oppath), and RaymondSmith (fpath_xrays).`

b. edit the file `preformal.f90`

change the parameter `'fpath'` (in module `preformal_var`) to the actual location (either relative or absolut) where you have saved the directory `DATA`

2. Decide whether you want to create Fastwind v10, or v11

To this end, set the parameter `OPTCMF_FULL` (in `nlte.f90, Module nlte_opt`) to `.False.` (v10) or `.True.` (v11). Moreover, set other important run-time options and parameters as well (see Part II for v10 and Part III for v11).

3. Edit the Makefile

with respect to

- the compile options of your local f90-compiler (in particular, the optimization),
- the default atomic model (explicit elements),

- and the `LINES**.dat` file (close to the end of the `Makefile`. The actual name will be written to `ATOM_FILE`, see below)

4. Choose a DETAIL data-set appropriate for your intentions.
Located in `inicalc`, following the `DETAIL` format. See file `detail.pdf`.

Examples are

• `A10HHe.dat`	H/He	hydrogen and helium, standard treatment of bound-bound collisions (CBB).
• `A10H__.dat`	only H	
• `A11HHe.dat`	as A10HHe, but with different CBB rates for hydrogen (following Przybilla and Butler 2004). Especially suited for IR work.	
• `A10HHeSi.dat`	rather old Si-Atom included	
• `A10HHeN2345.dat`	Nitrogen atom included, see Rivero Gonzalez et al. 2011/2012	
• `A10HHeCnewONSi.dat`	as above, but additionally C (unpublished) and O (from the WM-basic data base)	

Don't forget to update the default metallicity!
Change the X/H value close to the keyword ATOM

- IF no abundances for the explicit ions are provided in the input file INDAT.DAT, the values from `A*.dat` will be used!

- Only for He, INDAT.DAT always asks for a value, so that the He abundance in the detail input file does not need to be changed. **If** there is a specific abundance in `INDAT.DAT` for an element which is not treated as an explicit element, only the corresponding back-ground abundance is changed. See also below.

Note 1: the more elements are included, the longer the turnaround times**!**

Note 2: if you have modified the DETAIL file or created a new model atom with more ionization stages than currently present, please modify `subr. ilowimax_nlte` in `nlte_approx.f90` (middle part). For explicit elements, a maximum of 3 ions (globally) can be treated in parallel. Please take special care if you include ions that react strongly on X-rays (e.g., O VI). Do not forget to update the `LINES**.dat` file for new diagnostic lines!

5. Run 'make ATOM=file' or simply 'make',

where file is the name of your DETAIL data-set (see Sect. 4.), without the extension '.dat'

If you want to exchange your model-atom (i.e., you have already created executable objects from a specific detail file, and now want to use another detail file), first 'make clean'. This will delete all object and module files refering to the deprecated atomic model.

'make' will create the following executable objects:

- `pnlte_A*.eo` (for calculating the atmospheric structure and NLTE occupation numbers)
- `pformalsol_A*.eo` (for formal integral)
- `ptotout_A*.eo` (collects content of various data to further inspect the model structure)

* here refers to the name of the model atom, **without** extension '.dat' (e.g., `pnlte_A11HHe.eo`).

All the above objects have array dimensions derived from the DETAIL model atoms (in program `dimes.eo`), and written to `module nlte_dim` (in file `nlte_dim.f90`, created after `dimes.eo` was run, e.g., via the `Makefile`).

Note, that a number of additional objects (`.o`) and modules (`.mod`) are created.

For experienced users: of course, the `Makefile` allows for various additional commands and specifications. Please check!

E.g., `make clean` will delete all files created during complilation, except `ATOM_FILE` and the `*.eo` files.

WARNING!!! In case that you modify a `parameter` of the `.f90` programs (e.g., `fpath1`, `optcmf_full`), a simple new `make` is not sufficient to change this. Instead, before recompiling, `make clean`, and only then `make`.

Chapter I/IV. Running the `NLTE` code

To run the code, the input file `INDAT.DAT` must be present in your current *working directory*. Just after the program start, this file is written to the *model directory*.

In case you want to calculate without a consistent T-structure, or you want to use excellent start values for $T_e(r)$, you also need the files

`HOPFPARA_ALL_HHe and/or HOPFPARA_ALL_met`

in directory `../nlte`.

1. Structure and content of INDAT.DAT

Example 1

```
'cmfgengrid/s2a'                                line 1
T T        0       140                          line 2
0.0                                             line 3
44700.    3.79    19.6                          line 4
120.0     0.5                                   line 5
0.12E-04   0.1 2620.0   1.0   0.2               line 6
0.1       2.0                                   line 7
F F F T T                                       line 8
15.0      1.0     T   T                         line 9
T F        1        2                           line 10
1.0     0.1E-01  0.1                            line 11_a2
F                                               line 12_a1
0.76      0.4     12.41                         line 12_a2
N       7.92                                    line 13_a1
```

Example 2

```
'test                                           line 1

T T        0       100                          line 2
0.0                                             line 3
36250.      3.425      13.46                    line 4
120.0     0.6                                   line 5
0.354813E-06   0.1 1875.   1.25   0.1          line 6
0.12      2.0                                   line 7
F T F T T                                       line 8
15.     0.5     T  T                            line 9
T  F        1        2                          line 10
43.0    0.135     0.27                          line 11_a2
C       7.65                                    line 13_a1
N       8.55                                    line 13_a2
O       8.55                                    line 13_a3
SI      7.06                                    line 13_a4
P       5.12                                    line 13_a5
```

```
XRAYS   4.645474E-002                          line 13_b1
0.7500      25.00       1.500       591.0      -1000.  line 13_b2
```

In the following detailed description, the content of the various lines (not all required, depending on context) is provided by a typical example, displaying the variable names as used in the program(s). See also Examples 1 and 2 above.

line 1:

'A3030' — `modname`, (unique) name of *model directory* (where all model files will be stored), maximum 30 characters, in quotation marks

line 2:

T T n1 n2 — `optneupdate, helone, itlast, itmore`

`optneupdate`: usually T; if F, electron densities remain in LTE

`helone`: if T, He I will be calculated when He present
for Astars T (only dummy)
for Bstars T
for O stars T or F

at model start, n1= 0, otherwise
n1+1: first NLTE iteration when code (re-)starts
n2: number of iterations to be carried out
(typically 50 for v10 and 100 to 140 for v11)

important note: if performing some additional iterations (i.e., when re-starting an already existing model) then
in the case of Sobolev-treatment $n1 \geq 3$
in the case of CMF-treatment $n1 \geq 20$
in the case of CMFALL-treatment (v11) $n1 \geq 30$

line 3:

1. or 0. — `optmixed` = true (1.) or false (0.) controls whether all (0.) or only most important (1.) lines (from explicit elements) are calculated in CMF. Currently, set this number always to 0.

line 4:

T_{eff} log g R/R_{sun} — `teff, ggrav, srnom`
real numbers, all quantities defined at $\tau_{Lucy} = 2/3$ (spherical generalization of Rosseland optical depth)

line 5:

R_{max}/R_{star} T_{min}/T_{eff} — `rmax, tmin`, usually 120. and 0.6 (the latter value is the minimum temp. for the **initial** estimate of the T-stratification, but see above)

line 6:

M_{dot} v_{min} v_{inf} β v_{div} — `xmloss, vmin, vmax, beta, vdiv`, real numbers:

M_{dot} in M_{sun}/a
v_{min}: choose 0.1 (km/s) as start value
v_{inf}: in km/s
β: velocity field exponent
v_{div} : division between photospheric and wind regime; usually, 0.1 (in units of v_{sound}); should lead to a smooth transition at the sonic point, check by plotting v(r) or rho(r).
Note that in certain cases the input value of v_{div} does **not** comply with the value actually used, namely if a smooth transition is not possible. If such a case happens, it is logged in `FASTWIND.LOG`

line 7:
N(He)/N(H) XIHE `yhein, hei`
`yhein` atmospheric helium abundance, in units of hydrogen particle density
`hei` no. of electrons provided by helium ion,
start value for models = 0.0 (A-stars), 1.0 (B-stars), 2.0 (O-stars)

line 8:
F F F T T `optmod, optlucy, megas, accel, optcmf1`
`optmod` = F: photospheric model will be created by Fastwind
=T: external model will be read in from *working directory*.
Using an external model implies that `update_struct` will be set to F.
Currently, there are three possibilities:
`modname.michel` → wind model from Michel Cure,
`modname.struct` → photospheric model from Kurucz
`modname.tlusty` → photospheric model from Hubeny

where `modname` is the name of the *model directory*.

For file-structures, see `subr. modvl` in `nlte.f90`.
The first of the above external models connects a Fastwind photosphere with a Cure wind model, where the temperature structure can be calculated consistently or from `tlucy`. The 2nd and the 3rd model include a T-structure, and are connected with a Fastwind wind at `vdiv`. The external temp. is interpolated in `tlucy`, thus set `optlucy` = T.

`optlucy`=F: NLTE Hopf function (Santolaya-Rey et al. 1997) used, Hopf parameters required
=T: modified Lucy-approx. (Lucy 1971) for temperature

used. If you like to calculate a consistent T-structure (`enactor = T`) and have no Hopf parameters for the first temperature estimate available, this is the method to use. The "only" error will be a slightly erroneous density structure in the outer photosphere

`megas` = F: default
= T: file `megasalida` generated in *model directory*, displaying NLTE rates of all transitions of explicit elements (calculated in `subr. netmat` and `linesob)`

`accel` = T: default; forward extrapolation of n_i if possible **and** if `.not. almost_converged,` to accelerate convergence
= F: if acceleration does not work

`optcmf1`=T: CMF-treatment after iteration #20
= F: only Sobolev treatment, starting after iteration #3

line 9:
15. 1.0 T T `vturb, metallicity, lines, lines_in_model`
`vturb` micro-turbulence in km/s
`metallicity` in solar abundance units.
`metallicity lines lines_in_model`: 0. F F → no metallic background, only for v10 possible (should be cor rect, since no metallic line blanketing)

`metallicity` > 0: metal abundances (only non-specified elements), are scaled to the input value with respect to the **input** (file `abundan_solar`) solar **mass** fractions; e.g., 0.2 means that the (total) metallicity corresponds to two tenths of the corresponding total solar metal mass fraction (relative abundances remain as solar)

`metallicity` < 0: metal abundances (only non-specified elements) are scaled to the input value (abs) with respect to the **input** (file `abundan_solar`) solar **number** fractions; -0.2, e.g., means that each new abundance corresponds to two tenth of the original solar number fraction.

Note: As long as the metallicity mass fraction is small, the difference in the resulting abundances of both approaches from above is also small, as long as the He-content is solar! In other cases, the author suggests to use a positive value, however note that this is only a compromise!

It might be useful to play around with some numbers and study the output carefully:

The complete abundance output (in number and mass fractions) is logged in file `ABUNDAN`, which is created inside the *model directory* almost immediately after the start of the code.

`lines` = T: with approximate line-blanketing/blocking in v10

`lines_in_model` = T: approximate line-blanketing/blocking also in atmospheric model (→ τ_{Ross}, consistent lower boundary)
= F: only bf/ff metallic opacities in model; only for specific tests

line 10:

T F 1 2 `enatcor, expansion, set_first, set_step`

`enatcor` = T: T-correction scheme enabled; if set to F, implies that `update_struct` = F.

`expansion` = T: adiabatic cooling included, to be implemented in future versions, not working thus far (set to F)

`set_first` iteration number where corr. scheme shall start (suggested value = 1, beginning with Sobolev approx.)

`set_step` each `set_step` iteration the temperature will be updated (suggested value = 2, use 3 in problematic cases).

Note: Don't forget to check the variable `tmin1` in module `nlte.f90/fastwind_params`, whether it fits your problem.

Line 11_a1:

THICK `optthick` → T, only for v10. Optically thick clumping thus far not implemented in v11. If optically thin clumping aimed at (both v10 and v11), skip this line

Line 11_a2: Clumping parameters (up to 10), either for optically thin or thick clumping. Both for optically thin and thick clumping, the number of paramters tells which kind of clumping law to use.

For details, see `subr. clumping`. If optically thick clumping, three follow-up lines – 11_a3 to 11_a5 -- required (again with up to 10 parameters, see below).

Example for opt. thin clumping with linear law (for 3 given parameters)

10. 0.05 0.5 `clf,vclstart,vclmax`

The stratification of the clumping factor f_{cl} is assumed to be linear, starting at `vclstart` and reaching its maximum value at `vclmax`. Both velocities are in units of `vinf`. Inside from `vclstart`, no clumping shall be present (f_{cl} =1), and outside from `vclmax,` the clumping factor remains at its maximum value.

`clf` is this maximum value, which also corresponds (in the optically thin approach with only two components: clumps and interclump medium) to the *enhancement* factor. `clf` = 2, e.g., means that the density in the clumps is twice as high than the given M_{dot} would imply. In the 3-parameter approach, the interclump medium is always assumed to be void. For ρ^2 mass-loss diagnostics, corresponding processes remain almost unaffected as long as $M_{dot}\sqrt{f_{cl}}$ remains conserved.

To calculate *unclumped* models, set these parameters, e.g., to

clf=1., vclstart= 0.1 and vclmax = 0.2

To calculate a model with an (almost) constant clumping factor in the wind, set, e.g.,

clf=9., vclstart = 0.05 and vclmax = 0.1

All ρ^2 diagnostics corresponds then roughly to an unclumped M_{dot} of three times the input value (for more details, see literature, e.g., Puls et al. 2008, and `subroutine clumping`).

Note that the user can also apply other or own clumping laws, by either using more than 3 clumping parameters (e.g., the “Hillier law”, or the law by Najarro et al. 2011), or by manipulating the `subroutine clumping` in `nlte.f90`. Up to 10 input parameters are possible.

Line 11_a3:
Line 11_a4:
Line 11_a5:

If optically thick clumping shall be used within v10, use
line 11_a1 (keyword `THICK`)
together with a set of clumping parameters in
line 11_a2 (`clf_field`, clumping parameters),
line 11_a3 (`fic_field`, density contrast),
line 11_a4 (`fvel_field`, velocity porosity – lines), and
line 11_a5 (`hpor_field`, porosity – continuum).
The number of parameters (stratification) in all four lines needs to be identical.

For further details, see `subroutine clumping` in `nlte.f90` and Sundqvist et al. 2018.

Line 12_a1: Only required if `optlucy = F`. Otherwise, skip line

`T OR F` `hopfself;` If `TRUE`, Hopf parameters are taken from file `../HOPFPARA_ALL_xxx`, where `xxx=HHe` for abs(metallicity < 0.2) and `xxx=met` else (actually, for solar conditions).

Note, that this option currently requires that T_{eff}, log g and Y_{HE} lie **exactly** on the specified grid.

If `hopfself = F`, one has to provide an "extra" line with (three) Hopf parameters

Line 12_a2: Only required if `optlucy = F AND hopfself = F` (previous line). Otherwise, skip.

0.75 0.44 4.4 q_{inf} q_0 `gamma`

These parameters are read and used in `subroutine tlucy`. The equation applied for the determination of the NLTE Hopf function in plane-parallel geometry (see Santolaya-Rey et al. 1997) is

$$q(tau) = q_{inf} - (q_{inf} - q_0) * \exp(-gamma * tau)$$

where the definition of the parameters can be found in the IDL-routine `qhopf.pro`.

Line 13_a1…13_an: with n < 30
these lines can, but do not need to be present. Using the following specifications, one avoids to modify the `Detail` input file. Elemental names to be specified as in file `abundan_solar`, only capitals!

C 9.0
N 8.0 positive
SI 7.0

input **relative to hydrogen** (particle density) and always logarithmically, either with respect to H=12 (positive number) …

or

C -3.0
N -4.0 negative
SI -5.0

… or directly, log10(abund/H), but only if negative.

if log10(abund/H) > 0 (often when $Y_{He} > 1$), use only first possibility (with respect to H=12), since positive numbers always interpreted in this way

These lines (if present) refer to the prespecified abundances (explicit **and** background elements), **except for** He, which always has to be specifed in line 7.

If an explicit element has been specified, the corresponding abundance from the `Detail` input file will be overwritten (check!!!). If the specified element is from the background, a rescaled solar abundance will be overwritten.

Note that specified numbers will be used directly, i.e., **not** renormalized in `subr. abundan!`

Further examples

SI -4.00 means that SI (both as explicit and/or background element) shall be used with an abundance (relative to H) of 1.e-4

SI 8.0 has the same meaning.

Line 13_b1:

`XRAYS` 2.5e-2 keyword and f_X. If keyword is present, the following quantities (this and next line) describe important parameters of wind-embedded shocks that emit X-rays, which are partly absorbed by the cool wind. Only "allowed" for $T_{eff} > 25{,}000$ K and a metallicity $\neq 0$.

If keyword is absent, end of `INDAT.DAT` file.
2nd quantity in this line is either X-ray volume filling factor f_X as defined in Carneiro et al., 2016, Eq. (1), or quantity n_{so} for radius-dependent effective filling factor. See below.

Line 13_b2: if keyword XRAYS present in line before.

0.5 20.0 1.5 400. -1000 `gamx, mx, rminx, uinfx, px`
`For px` < -999.99, see Carneiro et al., 2016, Eqs. (7/8):
standard approach with constant filling factor, giving rise to ρ^2 dependent emission for both radiative and adiabatic shocks, see Feldmeier et al. (1997).

0.5 20.0 1.5 400. 1.5 `gamx, mx, rminx, uinfx, px`
`For px` ge -999.99, see Puls et al 2020 (Sect. 4, based on Owocki et al. 2013): alternative approach with radius-dependent filling factor, giving rise to ρ-dependent emission for radiative and ρ^2 dependent emission for adiabatic shocks. The actual filling factor is calculated from the variables f_X, `(rminx, mx)` and `px`, see Eq. 12 in Puls et al. 2020 that define, together with Eq. 8 and other shock-related quantities, the effective volume filling factor. Here, the input quantity f_x corresponds to the quantity n_{so} in Eq. 12.

Note: in both cases, the variable `mx` is required as well, since the effective minimum emission radius is calculated from
min*(rminx, mx*vsound)*

2. Running the code

As soon as all files are prepared (Chap. I/III) and `INDAT.DAT` has been specified, one can run the code by

```
pnlte_A*.eo
```

or

```
pnlte_A*.eo >filename
```

(where, as above, * needs to be replaced by the actual name of the atomic data file (explicit elements)).

For the 2nd command, the output that normally appears on the screen is written into a file named `filename`. Note already here that a condensed form of this output is always written to the file `FASTWIND.LOG` in the corresponding *model directory*. The content of this file has changed substantially from versions v10.8/v11.8 on.

If

`n1=0  n2=0`

in line 2 of file `INDAT.DAT`, then only an estimate of the hydrodynamic model atmosphere is generated, together with one LTE iteration.

Otherwise the code proceeds from iteration `n1` to `n1+n2` and generates results and output.

If the corrections fall below a certain threshold, then

a final iteration is calculated, and indicated by

`ESTO ES EL ACABOSE !` (guess or inform yourself what this means).

This statement also appears if the model has not converged, but reached the maximum number of iterations. If this statement does not show up, there should be a message about an error that has appeared (both in the running output and in `FASTWIND.LOG`).

IMPORTANT NOTE: if you want to restart a model with **converged** T-structure (calculated with `ENATCOR = TRUE`) and want to force the code to perform additional T-corrections, you need to delete the file `MAXTCORR.dat` manually.

3. Has the calculation converged?

The most simple check on the convergence is to check whether the number of actually performed iterations (→ file `CONVERG` or `CONVERG_METALS`, or `FASTWIND.LOG`) is lower than the maximum number of iterations constrained in `INDAT.DAT` (`itmore`, line 2), **and** that the code has stopped with the above message `ESTO ES EL ACABOSE.`

In case that the number of performed iterations is lower than `itmore`, but the code has stopped with a different statement or crashed, the situation is more difficult. If the code has stopped with a specific statement, one might try to find out what has been going on (maybe by means of the code itself), and to improve the situation (e.g., by changing parameters and/or options; often the comment given by the stop statement will be somewhat helpful). If the code has crashed, one might check whether there was an error in the usage (wrong input file(s), wrong atomic data etc.), and, if not, try to improve the situation by a new run with one or more slightly changed input parameters.

The most difficult situation, however, occurs when the code stops with "`acabose`", but has reached the maximum number of iterations without converging. In this case, please consult the `FASTWIND.LOG` file for the last iteration, and check which of the levels at which depth points prevent convergence ("`CORR. MAX:`", with a number larger than 0.003). If only one depth point is affected, and the responsible level is an upper one (not the ground or a metastable level), one might consider the model as basically converged. Otherwise (if more than one depth point is affected, and/or the corresponding level is an important one), one might try to continue the iteration cycle, by setting `itlast` to the previous value of `itmore`, `itmore` to roughly 100, and re-

starting the program. It is highly likely that this additional run will lead to convergence. If not, then there might be physical problems that prevent convergence.

Chapter I/V. NLTE model output

`pnlte_A*.eo` generates the files enumerated below in the *model directory*. Please remember: if you have specified new clumping parameters and/or new abundances, please interrupt the code after one or two seconds, and inspect the files `ABUNDAN` and/or `CLUMPING_OUTPUT` in the *model directory*, whether everything has been initiated as desired. If so, run the code until its end. Otherwise, modify the input.

v10: in dependence of the number of explicit atoms, the wall clock time is (on a typical notebook) 10 to 50 minutes (incl. formal solution, see below).

v11: again in dependence of the number of explicit atoms, the wall clock time is one to two hours (incl. formal solution, see below).

IMPORTANT NOTE: in addition to the files below, a number of scratch files are generated in the `tmp` directory. Be careful to set the relevant environment variable (usually `TMPDIR`) to a local disk to avoid any delay due to net-activity.

For further analyses, please note:

n(k, j) is the particle density for element number k and ionization stage j (astron. convention)

in file `FLUXCONT`:

`TRAD`= radiation temperature related to inner radius
`TRAD1`= rad. temperature related to tau'=2/3 (for def. of tau', see Santolaya-Rey et al. 1997), usually the temperature one is looking for.

`RTHIN`= radius from which $H_\nu(r) \propto 1/r^2$
`LTHIN`= depth-index of `RTHIN`

All variables depend on frequency, of course.

Files for v10 and v11

name	type	content
FASTWIND.LOG	ASCII	program versions, important options, abundances (explicit and background), important output, warnings, error messages, mostly related to stop statements.

ABUNDAN	ASCII	abundances for background (and explicit) elements, both by number and by mass
ADD_FREQUENCIES	ASCII	additional freq. points to increase resolution
BG_LINES. CMF	ASCII	important lines from background elements calculated in CMF (single line)
CHIBAR_H	ASCII	flux-weighted mean opacity/mass and g_{Rad} (using flux-weighted mean and nominal flux), from radiation field using pseudo continua (see introduction). Only photo-spheric values are of correct order. Note that until May 2026, the provided g_{Rad} was actual-ly (r/SR)^2 g_{Rad}, with SR the radius value of the lower most point.
CLUMPING_OUTPUT	ASCII	clumping parameters as a function of depth, for both thin and thick clumping
CONT_FORMAL	binary	opacities, emissivities for (pseudo) cont. quantities
CONT_FORMAL_ALL	binary	opacities, emissivities, J_{nu}, Thomson-parameter
CONVERG	ASCII	convergence properties (explicit elements)
CONVERG_METALS	ASCII	convergence properties of metal background
DELTA_TC_LA.dat	ASCII	temperature structure for most recent temp. iteration
DERIVADIFF_TC_LA.dat	ASCII	important derivatives (differences) for most recent temp. iteration
DERIVATIVES_TC_LA.dat	ASCII	important derivatives for most recent temp. iteration
DIFF_TC_LA.dat	ASCII	temperature structure for most recent temp. iteration (*this is the most interesting file* related to temperatures)
ENERDIFF_TC_LA.dat	ASCII	heating/cooling energy differences for most recent temp. iteration
ENERGY_TC_LA.dat	ASCII	heating and cooling energies for most recent temp. iteration
ENION	binary	NLTE ionization fractions explicit elements, n_e (NLTE)
FLUXCONT	ASCII	F_ν, TRAD, TRAD1, RTHIN, flux conservation
GRAD.OUT	ASCII	estimate of photospheric (r/SR) ^2 g_{Rad} after correction of photospheric structure, otherwise start values
GRAD_START.OUT	ASCII	estimate of photospheric (r/SR)^2 g_{Rad}, start values

INDAT.DAT	ASCII	(almost) copy from actual input file
INXRAY.DAT	ASCII	either 'NO XRAY TREATMENT', or shock parameters
LTE_POP	binary	LTE occupation numbers for explicit elements
MAX_CHANGE	ASCII	convergence properties of explicit elements
MAX_TCORR.dat	ASCII	convergence properties of temp. correction
METAL_IDL	ASCII	for `ndebug` ge 2, ionization fractions of background (metals ne 0)
METAL_IDL_LTE	ASCII	for `ndebug` ge 2, ionization fractions (LTE) of background (metals ne 0)
METAL_IONS	ASCII	ionization fractions of metals (metals ne 0), output in WM-basic format
METAL_IONS_LTE	ASCII	ionization fractions (LTE) of metals (metals ne 0), output in WM-basic format
METAL_RESTART	binary	data required for restart (metals ne 0)
MODEL	binary	atmospheric structure
NLTE_POP	binary	NLTE occupation numbers for explicit elements
OPA_EFF_RAT_LA	ASCII	info on optically thick clumping (radius and frequency)
REL_ENERGY_TC_LA.dat	ASCII	relative heating and cooling energies for most recent temp. iteration
TAU_ROS	ASCII	τ_{Ross} (v10: LTE; v11: NLTE), and n_e(LTE)
TEMP	binary	temperature stratification

Additional files for v11

CHIBAR_H_CMF	ASCII	flux-weighted mean opacity and g_{Rad} for complete frequency range, calculated in CMF (within `wavblue` and `wavred`), and approximate (outside that range). g_{Rad} calculated from mean opacity and nominal flux.
CONT_FORMAL_CMF	binary	various cont. quantities in CMF (most important, NO _LINES), for formal solution
KCMF	ASCII	index of start and end of CMF treatment

LINE_LIST_MERGED	binary	line list for background and explicit elements. NOTE: level indices for explicit and bg elements have different structure
OCCNG	binary	met_imin, met_imax, occng for CMF-restart
OCC_NO	ASCII	important N and O resonance lines (opa and eta)
out_rad_force	ASCII	output for radiative acceleration (various components), current iteration
out_xj_app, out_xj_cmf, out_xj_cmf_coarse , out_xj_smoothed, out_xj_saved, out_xh_obs,		
out_xh_obs_coarse	ASCII	J_ν and H_ν in obs. frame and CMF, high resolution and coarse frequency grid. Calculated if `optout_xj_xh = T`

Chapter I/VI. Analyzing the NLTE model output

To generate a concise file containing most of the important output, run

```
ptotout_A*.eo
```

("automatically" compiled/linked when running `make`)

The program requires (asks for) the name of the *model directory*, and after less than one second turnaround time, the output file `OUT_TOT` will have been created in that directory.

To visualize various quantities, you can use either IDL (not for free), GDL (for free), or translate the IDL-routines (in directory `NLTE/idlproc`) into any language of personal convienence. In the following, we describe the approach using IDL/GDL. Note: some routines might need to be adapted, since used by the author for specific work.

Before starting with IDL/GDL, the initialization files `startup.pro` or `startup_gdl.pro` need to be adapted **once**, and the path towards these files to be specified in the resource file of the user's shell (e.g., in `.bashrc`). Please note that all directories and subdirectories containing procedures that will be used need to be included into the `!path` variable defined in the startup file(s). For details, search the internet or use AI. In the following, we will use IDL as an alias for IDL/GDL.

In order to inspect a specified atmospheric model, the above created `OUT_TOT` file needs to be read by the (IDL) procedure/script

```
readmod.pro
```

via the command

```
readmod,'obstars/star1/OUT_TOT','name',shortname
```

where `obstars` is the (path to the) current *working directory*, and `star1` the specific *model directory*. In this way, a variety of IDL arrays (describing the model atmosphere and few radiative quantities) are created, that can be used and displayed by several IDL routines.

`name` will appear on plots as header.

In particular, the arrays are

`shortname.r` `.v` `.dvdr` `.rho` `.p` `.t` `.xne` `.taur` `.m` `.levnames` (level names)

`.dep` `.ifrac` `.lam` `.fnue` `.trad` (corresponds to TRAD1) `.hnue` `.fcl` (clumping factor)

The routine assumes that the maximum radius is 120 Rstar. If this is not the case, please check the routine and provide the modified maximum radius via the input (required for correct fluxes!).

1. Plot of model structure

suppose that two (different) models have shortnames a1 and a2. Then the IDL commands

```
plv,a1    plrho,a1    plt,a1    plxne,a1    plp,a1    plifrac,a1
```

will produce plots with header 'name' of these arrays as a function of

either r, taur (plane-parallel Rossland optical depth) or m (plane-parallel column density). The routines will ask you about your preferred x-axis.

You can always overplot by, e.g.

```
oplot, a2.taur, a2.v
oplot, a2.r, a2.v
```

etc.

Instead of these plotting procedures, you can of course also use the usual

```
plot, a1.taur, a1.v
```

but then you will plot only the standard display without titles and headers.

2. Plot of ionization fractions (H, He, metals, explicit and background)

For plotting the fraction of explicit elements, use `plifrac,a1`

where here and in the following, `a1` shall be an example for `shortname`.

If the file `METAL_IDL` is present (requires `ndebug=2` (default) in `module fastwind_params`), the ionization fractions of the (approximate) background metal ions can be plotted as well, in analogy to `plifrac`; the command, however, is

```
plifrac_met,cat,a1
```

where `cat` is the *model directory*. Lots of other options allowed, inspect procedure.

3. Plot of source function

The line source function (in units of B_ν(TRAD)) is plotted by

```
plsic,a1
```

with

example for input lower level: 1 (IDL array convention, i.e., corresponds to n=2)

example for input upper level: 2 (n=3)

or 3 (n=4) etc., and input of wavelength (in Angstrom).

4. Plot of departure coefficients

use the routines

```
pldep,a1
```

```
pldepn,a1
```

5. Plot of flux

the flux distribution is plotted by

```
plflux,a1
oplot,a2.lam,a2.fnue
```

and a plot of the eddington flux (rescaled to R(τ=2/3)) results from

```
plhnue,a1
```

Chapter I/VII. Formal integral, and required data

For calculating the formal integral(s), four files need to be present/checked:

`pformalsol_A*.eo`	code for formal integral ("automatically" generated when running `make`)
`FORMAL_INPUT`	input file to calculate various line complexes and wavelength ranges with `pformalsol_A*.eo` (see below)
`thom_new.dat`	line broadening data for H/He (in `inicalc/DATA`)
`LINES**.dat`	line data for metal lines (as specified in `Makefile`, last part, located in `inicalc/DATA`)

1. ATOM_FILE

A fifth file named `ATOM_FILE` is automatically generated within `Makefile` (whenever the `DETAIL` atomic data file A*.dat has been changed). It contains the names of the detail file and of the files for the broadening data – three lines in total.

`ATOM_FILE` is located in the current *working directory* (i.e., in parallel to `INDAT.DAT`, see Fig. 1)

Example for `ATOM_FILE`

```
A10HHeN2345.dat
thom_new.dat
LINES_CnewNOSi_new_coll.dat
```

The file from the 1st line needs to be copied/modified from `inicalc` into the current *working directory* (in parallel to `INDAT.DAT)`, the 2nd one is located in `inicalc/DATA` (modify only when you know what you are doing), and the 3rd one also in `inicalc/DATA`. This file (`LINES**.dat)` might need to be updated, when new lines shall be synthesized (see below).

In most cases, however, it will be sufficient to check and update the file

2. FORMAL_INPUT

For each line/line complex, the following info needs to be present:

1. name of line(complex). If name = UV (only v11, in the following denoted by 'UV-type'), then a normalized spectrum for a range of wavelengths will be synthesized, see item 6 below. UV-type spectral ranges are calculated always for vacuum wavelengths.

2. total number of considered lines

3. 1st component:
   a. lower level
   b. upper level
   c. transition number: 0 if transition from H or He (no entry in LINES**.dat)
      1… n for multiplets with same lower and upper level (packed), and broadening option 2 or 3; transition number must be consistent with corresponding entry in LINES**.dat.
   d. broadening option: 0 … 3, see below
4. 2nd component:
   a. lower level
   b. upper level
   c. transition number: 0 if transition from H or He (no entry in LINES**.dat)
      1… n for multiplets with same lower and upper level (packed), and broadening option 2 or 3; transition number must be consistent with corresponding entry in LINES**.dat.
   d. broadening option: 0 … 3, see below
5. 3rd component …
6. If name = UV, then lower and upper wavelength limit of considered range, and resolution.

3. Broadening options

0 … pure Doppler broadening

1… Broadening for H/He (mostly linear Stark, but also BCS, ISO and ISO1 approach), from tables. In case of linear Stark and if no table entry is found, broadening either from Griem approx., or from Lemke 1997 (for hydrogen with upper level $n_u \leq 10$), see below.

2… Voigt broadening (includes quadratic Stark) from LINES**.dat, with corresponding radiative and collisional damping.

3… Voigt broadening with approximate damping parameters, works only for one comp., see `preformal.f90`. All values are calculated from the specified levels and atomic model

Examples:

HEII4686a 5 HE23 HE24 0 1 HE12P3 HE14S3 0 1
N310 N311 1 2 N310 N311 2 2 N310 N311 3 2
UV 900 2000 0.1
UV 3800 7500 0.05

For transitions between very high lying levels, there is the possibility to obtain an LTE approximation: for this path, set broadening option to 0 or 1, and ncomp = 1(!)

Note: all wavelengths of individual lines/line complexes are corrected for terrestrial refraction (standard conditions) if wavelengths > 2000 Å, otherwise vacuum wavelengths. Use routine `refrac.pro` to convert from vacuum to air, and `refrac_inv.pro` vice versa.

Some useful level designations for HeI/II (e.g., from A10HHe) are

HeI4026 HEI2P3 HEI5D3 blended with HE24 HE213

HeI4387 HEI2P1 HEI5D1

HeI4471 HEI2P3 HEI4D3

HeI4922 HEI2P1 HEI4D1

HeII 4200 HE24 HE211

HeII 4541 HE24 HE29

HeII 4686 HE23 HE24 (in case, consider blend with HeI and NIII)

4. Broadening of H and He

- In addition to the H and He broadening as derived/tabulated by Schöning and Butler 1989 (incl. the ISO/ISO1 approach for He I lines, see file `inicalc/DATA/thom_new.dat`), …

- … **other** H/He lines (particularly in the IR) can be treated *either* within the Griem approximation (Griem 1967) following Auer and Mihalas (1972, Appendix); this is the default for all transitions not covered above and except for Paschen and Brackett lines with upper level $n_u \leq 10$,

- … *or* using the results for hydrogen from Lemke (1997, until upper level n=10 – transitions to higher upper levels are affected by numerical problems, see Repolust et al. 2005), tabulated in files `balmer_lemke.dat`, `paschen_lemke.dat` and `brackett_lemke.dat` in `inicalc/DATA`). This is the default for Paschen and Brackett lines with upper level $n_u \leq 10$. Lemke-broadening for other lines (experimental) can be

switched on by changing the corresponding logical variables in `formalsol.f90` and `preformal.f90` (search for "lemke", change the maximum upper level, and check subroutines `vcs_balmer` and `vcs_pb`).

5. Running the code

`pformalsol_A*.eo` requires three input lines:

1. `Name` of *model directory*
2. microturbulence
   a. photospheric microturbulence, one value, in km/s , or
   b. variable v_{mic}, two values, min(v_{mic}) in km/s and max(v_{mic}) in km/s or in units of v_{inf}. In between, the code performs a linear interpolation as a function of $v(r)/v_{inf}$.
3. incoherent electron scattering: with (=1) or without (=0) incoherent electron-scattering (accounting for correct thermal broadening of electrons) following Rybicki & Hummer (1994). Can be currently applied only in the optical (> 3000 A), and for non UV-type (i.e., only for lines/line complexes, not ranges). Should also work for optically thick clumping, but needs to be tested. Relevant only for emission lines such as Balmer-alpha or HeII 4686.

… and generates one or two files per treated line complex in the *model directory*,

`IX.DAT` and

`STARK.BIN` if line is from H or He.

These files are always created and the old ones overwritten before a new line complex will be considered. `STARK.BIN` contains the broadening functions for H/He (if not tabulated, in Griem approximation or from Lemke), and `IX.DAT` the following info if the profile type is **not** `UV`:

1. number of lines in complex
2. for each line in complex
   a. wavelength (in vacuum or air, see above)
   b. statistical weights (lower and upper level)
   c. f-value
   d. lower and upper level, by index (explicit element)

e. atomic weight

f. broadening option

g. if Voigt broadening (option = 2 or 3), collisional and radiative damping parameters, from `LINES**.dat` (see below).

If the profile type is `UV` (only in v11), then Stark broadening for H/He, but pure Doppler broadening for other elements (explicit and background). In this case, the files `IX.DAT` and `STARK.BIN` refer only to H/He lines within the considered range, and all wavelengths are in vacuum.

The file LINES**.dat

Broadening data for explicit elements different from H/He (broadening options 2 and 3) are provided in the files `LINES**.dat` (`inicalc/DATA`), and might need to be updated, whenever new lines (non H/He species) shall be synthesized, or if packed explicit levels need to be de-packed within the CMF radiative transfer (e.g., the N III resonance lines in the EUV if nitrogen is an explicit element).

In file `LINES**.dat`, there is the following info:

For each ion, there is a DETAIL-like header before the actual data begin,

CL TYPE RBB 2 7 RBB

and

0

at the end of the specific ion.

Note that the above specification is different from the standard `DETAIL` input! Basically, it tells that the following entries are RBB (radiative bound-bound) transitions, described by 7 parameters (starting with wavelength).

Example for NIII resonance lines and potential ‘f’-lines in the optical

```
CL  TYPE   RBB  2   7  RBB
N31    N311    1   374.198  0.1383  1.07E+09      0.    0.   7.09E-09       6
N31    N311    2   374.434  0.25     1.07E+09      0.    0.   7.09E-09       6
N31    N311    3   374.441  0.0277  1.07E+09      0.    0.   7.09E-09       6
N310   N311   1  4634.122  0.1367     9.79E8      0.    0.   2.14E-08       6.
N310   N311   2  4640.641  0.2454     9.79E8      0.    0.   2.14E-08       6.
N310   N311   3  4641.850  0.0271     9.79E8      0.    0.   2.14E-08       6.
….
0
```

Content of columns 1 to 10 (see also header of `LINES**.dat` file)

column 1: lower level (explicit element)
column 2: upper level (explicit element)
column 3: number of component within multiplet (arbitrary, but unique and consistent with transition number in `FORMAL_INPUT`, see Chap. I/VII/2).
column 4: wavelength (in vacuum or in air, see above)
column 5: oscillator strength provided as $g_i f_i$/Sum(g_i), where Sum(g_i) is the statistical weight of the lower, **packed** level, and $g_i f_i$ the gf-value of the individual transition within the packed multiplet (see Sect. "Level packing" at the begin of this manual). **No** internal cross check is performed, i.e., if this quantity is wrong, then everything is wrong.
If there is only one component (physically, or because there is no packing), then this quantity is the actual oscillator strength.
column 6: radiative damping constant (=1/lifetime) (sum of all Einstein A-coefficients that connect the lower and upper level of the considered transition with all possible levels below that lower and upper level), divided by 4 pi.
column 7: 0. (historical reasons)
column 8: 0. (historical reasons)
column 9: collisional damping constant, divided by 4pi, needs to be multiplied by electron density
column 10: contains Sum(g_i) (lower level, see above), and has to be consistent with the atomic data file; **no** internal cross check performed.

In most cases, radiative lifetimes from NIST/VALD databases, and collisional damping either from VALD, or according to Cowley (1971), or mostly following Kurucz (see `LINES**.dat` files)

All synthesized lines/line complexes as requested in `FORMAL_INPUT` are stored within the *model directory* under the appropriate `name` and an ending describing the value of v_{mic} and the treatment of electron scattering (with incoherent scattering → `esc`). In case of no ending, `escat = false` and `vturb = 0`. has been adopted.

Examples:

`OUT.HALPHA_ESC_VT015`	Halpha complex with incoherent electron scattering and constant v_{mic}
`OUT.HBETA_VT015`	
`OUT.HGAMMA`	`escat = F` and v_{mic} = 0.
`OUT.NIII4511_VT010`	
`OUT.NV1240_VTV015`	`VTV015` variable v_{mic}, here with minimum 15 km/s
`OUT.UV_00900_02000__0.100_VT015`	spectrum in range between 900 and 2000 Å, with resolution 0.1 Å, and v_{mic} = 15 km/s

Chapter I/VIII. Plotting/inspecting the line profiles and spectral ranges

All provided plotting routines are written in IDL/GDL. In the following, we will discuss only few of them, and concentrate on the most important features. For details, please inspect the individual routines. Note: some routines might need to be adapted, since used by the author for specific work.

1. Plotting individual line profiles

… via `halpha.pro`

```
halpha,'astars/a1234/OUT.HALPHA_VT005'
halpha,'astars/a1234/OUT.HBETA_VT005'
halpha,'astars/a1234/OUT.HGAMMA_VT005'

halpha,'ostars/a1234/OUT.HEI4471_VT015'
halpha,'ostars/a1234/OUT.HEII4200_VT015'
```

etc.

`halpha,'astars/a1234/OUT.HALPHA_VT015,vsini=150` convolve with vsini

or

`halpha,'astars/a1234/OUT.HALPHA_VT015,x,y,vsini=150`

line complex exported to local arrays x and y

2. Comparison with observation

either

```
oplot,xobs,yobs
```

or

use keywords `wobs, pobs` in `halpha.pro`

3. Multiple line profiles from explicit elements

various optical hydrogen lines:	`hprof. pro`
various optical HeI lines:	`heiprof.pro`
various optical HeII lines:	`heiiprof.pro`
various optical H/He lines:	`plotall.pro`
various IR H/He lines:	`irprof.pro`

various optical NII to NV lines: `nprof.pro`

various SiII lines `si2prof.pro`

various SiIII lines `si3prof.pro`

etc.

Note: in case, change names of lines/line complexes and xrange

```
heiprof,'astars/a1234','title',ext='VT015',vsini=150.,vmac=50.,
resol=10000,ps='a1234.ps'
```

- output to ps-file
- convolution with projected rotational velocity via option `vsini` (in km/s)
- convolution with Gaussion via option `resol`.
- convolution with RT (radial-tangential) macro-turbulence (model by Gray) via option `vmacro` (in km/s)
- *avoid convolution with one of the above quantities being equal or very close to zero.*
- before convolution can be performed, at first compile procedure with `.run convol`
- comparison of spectra from several different models:
  ```
  plotall,['astars/a1234','astars/b1234','astars/c1234'],
     'title',ext=['VT015','ESC_VT010','VT010']
  ```
- comparison with observations possible, of course (look into path with `comp`)

4. Plotting a spectral range (v11)

See (and in case modify) `plot_range.pro`

Basic idea:

```
model1='ostars/A4040'
readcol,model1+'/OUT.UV_00900_02000__0.100_VT015',idum,lam1,
                xhdum1,prof1,xhdum1
bb=where(lam1 ge 850 and lam1 le 2100)
lam1=lam1(bb)
prof1=prof1(bb)
convol,lam1,prof1,l1,p1,vsini=150,vmac=50,resol=10000
```

```
lam1=l1
;in case (for optical ranges):
refrac,lam1
prof1=p1
!p.multi=[0,1,2]
window,0,xsize=1200,ysize=800
plot,lam1,prof1,xrange= [900,1500],xs=1,yrange=[0,2],ys=1
plot,lam1,prof1,xrange=[1400,2000],xs=1,yrange=[0,2],ys=1
```

Part II. Fastwind v10 specifics

In this section, we discuss few issues which are of concern for v10 alone.

II/1. Options and parameters

in modules `nlte_opt` and `fastwind_params`, respectively.

1. Options (in `nlte_opt`)
   `optcmf_full = .false.`
   `update_struct = .true.` (except for test cases or if problems)
   `opt_oiii_iter = .false.`

   if enhanced output for opacity project data required, set `opdebug = .true.`

2. Parameters (in `fastwind_params`)
   energy ranges for X-ray luminosities: modify `xred1` etc.
   output of partition functions for bg elements: set `ndebug = 3`

If `update_struct = .true.` (see above), the hydro-structure is iterated to obtain an (almost) consistent density-structure. In this case, the Hopf-parameters (for the initial estimate of the temperature structure) do no longer need to be chosen as carefully as without such update, and rough values are adequate. In most cases, it is sufficient to use Lucy's T-structure for the start model (2nd logical variable in line 8 of `INDAT.DAT` set to T).

II/2. First test calculations

As a first test run, use `A10HHe.dat` and the sample INDAT-file for A3030. Compare your results with the comparison model as calculated by the author, contained in `inicalc/A3030.tgz`, both with respect to structure and to synthetic spectra. Small differences will be visible, since the author's version has been calculated with an older version. Calculate a 2nd version of this model with background-elements data base `ATOMDAT_NEW_old`, and compare model-structure and line profiles with those from your original model.

If you would compare model A3030 with results obtained by using quite early program versions, you would see larger differences in He II, but the rest would be fine. The reason for these differences is the temperature and electron-density structure, which is different when not updating the hydro-structure. The largest differences would be found for dwarfs (or other weak-winded models) in the T_{eff}-range between 30 to 35 kK.

II/3. How to change the abundances of only a few metals

(if stellar/wind parameters and YHE do not change)

This can be accomplished by few additional iterations:

i) copy your old model to a new directory

ii) in the `INDAT.DAT` file,

- change the model name according to the new *model directory*
- set ITLAST to 20 (v10) or 30 (v11) or larger,
- change the abundances as you like

iii) restart the model, i.e., execute `pnlte_A*.eo`

iv) calculate, for the new model, `OUT_TOT` (`ptotout_A*.eo`), and the line profiles (`pformalsol_A*.eo`).

When a model will be restarted (`ITLAST` = 20 or 30) , the model structure (T, ρ) will remain unchanged (except you hack the program). In earlier program versions, also the metallic background (except for the continuum of the explicit elements) has been kept constant, and only the CMF/continuum iterations for the explicit elements were performed.

From v10.1 on, this has been changed: Now, if the changes in abundance are larger than 20%, the background is re-iterated as well, to consistently account for the *line-blocking* effects of the modified elements.

Anyway, since the model structure does not change (the effects of metal-abundance changes on T and ρ are discarded), and H/He are almost not affected, the convergence is fast, and for such a restarted model only order of 15 iterations are required (compared to a complete model, which needs 40 to 60 iterations for v10 and more than 100 iterations for v11)

In those cases where you change a multitude of metallic abundances, calculate a new model, since then the effect on T and ρ can no longer be neglected!

If you change YHE and restart the model, you will get a stop, since then the density would be significantly affected.

II/4. Minimum set of files to save

The minimum file set to save after having calculated a specific model is

FASTWIND.LOG
INDAT.DAT
INXRAY.DAT
OUT_TOT
OUT.* (profile complexes and spectral ranges)

If other checks such as for convergence and flux conservation should be possible, save additional files (see Chap. II/5 and III/4).

II/5. Model files required to rerun `formalsol`

In order to recalculate specific or to calculate additional line-profiles, certain files need to be present and *must not be deleted* when saving the model files. These are

FASTWIND.LOG
CLUMPING_OUTPUT
CONT_FORMAL
ENION
FLUXCONT
INDAT.DAT
INXRAY.DAT
LTE_POP
MODEL
NLTE_POP

Part III. Fastwind v11 specifics

In this section, we discuss few issues which are of concern for v11 alone.

III/1. Options and parameters

in modules `nlte_opt` and `fastwind_params,` respectively.

1. Options (in `nlte_opt`)
   `optcmf_full = .true.`
   `update_struct = .true.` (except for test cases or if problems)
   `opt_oiii_iter = .true.,` if N is included in the explicit elements and the optical NIII f features should be adequately predicted.
   Otherwise, set `opt_oiii_iter = .false.`

   if enhanced output for opacity project data required, set `opdebug = .true.`

2. Parameters (in `fastwind_params`)
   wavelength range of complete CMF treatment: `wavblue, wavred`
   (default: 200,10000 Å)
   energy ranges for X-ray luminosities: modify `xred1` etc.
   output of partition functions for bg elements: set `ndebug = 3`
   `optout_xj_xh` (default: `.true.`): allows for output of `out_xj_cmf` etc.

III/2. First test calculations

As a first test run, use `A10HHeN2345.dat` and the sample INDAT-file for s4a (a hot supergiant). Compare your results with the comparison model as calculated by the author, contained in `ini-calc/s4a.tgz`, both with respect to structure and to synthetic spectra. There should be (almost) no differences. Compare (for your results) the individual profile complexes with the UV-type calculations, by overplotting. Compare, for your results, the ionization fractions of nitrogen, when either treated as an explicit element (`plifrac.pro`) and when treated as a background element (`plifrac_met.pro`), again by overplotting. Calculate a 2nd s4a-version with background-elements data base `ATOMDAT_2016_new`, and compare model-structure, line profiles, and UV-type wavelength ranges with results from your original model.

Note: Thus far, calculation and convergence properties of v11 have only been tested with explicit elements H/He or H/He/N. If other metals should be used as explicit elements, please test carefully.

III/3. Minimum set of files to save

The minimum file set to save after having calculated a specific model is equal to the case of v10:

FASTWIND.LOG
INDAT.DAT
INXRAY.DAT
OUT_TOT
OUT.* (profile complexes and spectral ranges)

If other checks such as for convergence and flux conservation should be possible, save additional files (see Chap. II/5 and III/4).

III/4. Model files required to rerun `formalsol`

In order to recalculate specific or to calculate additional line-profiles (incl. the v11-only UV-type), certain files need to be present and must not be deleted when saving the model files. These are

FASTWIND.LOG
CLUMPING_OUTPUT
CONT_FORMAL

CONT_FORMAL_ALL
CONT_FORMAL_CMF
ENION
FLUXCONT
INDAT.DAT
INXRAY.DAT
LINE_LIST_MERGED
LTE_POP
MODEL
NLTE_POP
OCCNG

Compared to the required files for v10, these are additionally

CONT_FORMAL
CONT_FORMAL_ALL
CONT_FORMAL_CMF
LINE_LIST_MERGED
OCCNG

To Do List

In this section, we will briefly enumerate those issues which need to be improved or still need to be implemented into Fastwind. Since the main author has meanwhile retired, this needs to be done by other persons.

Regarding both v10 and v11 …

- Thus far, very dense winds such as those from WR stars (where τ_{Ross}=1 is reached already well in the wind) cannot be treated. This is **no** principal problem, and after future, careful testing, removing specific stop statements, and adapting few current approaches, one should be able to solve this problem.

- At least for tests, the temperature stratification might need to account for adiabatic cooling (because of wind expansion). However, this has an effect only in the outer wind, where the impact of the temperature is anyhow almost negligible except for very dense winds.

- At some point, level dissolution (see Hubeny et al. 1994 and Hillier and Miller 1998) needs to be incorporated, particularly for a precise description of IR lines. The impact on UV and optical lines is quite low.

- As already stated at the beginning of this manual, a more precise description of iron-like elements with a multitude of levels requires the implementation of a super-level approach.

This should be fairly easy for background elements, but for *explicit ones* additional considerations (mostly regarding the CMF radiative transfer in `nlte.f90`, now for depacked lines) might be necessary: **for v10**, depacked data plus a suited depacking algorithm would be required, since explicit levels are currently depacked only in `formalsol`. **In v11,** depacked data for explicit elements (from `LINES**.dat`) are already now merged with background data in `subr. line_list (cmf_all.f90)`, and the algorithm needs only to be improved regarding a speed-up of the process, because of the multitude (thousands!) of depacked line components.

- In the latter case (and also in other situations), more than **a total** of 999 (packed) explicit levels might be required, and a different method of indexing these levels (which currently is limited to such 999 levels) might need to be incorporated. As well, the current limit of (packed) background element levels is given by 99 levels **per ion**[5], and the indexing would need to be changed if this number were no longer sufficient (quite unlikely, however).

- At least for iron, the ionization stages II, III (and series 4f from IV) need to be improved (mostly by including higher-lying levels), particularly for transitions located longward of the UV, and independent whether the current packing philosophy is kept or replaced by the super-level approach.

  In v10, such an update would need to be performed for iron *as a background element*, since the atmospheric structure is calculated from the *complete*[6] metallic background, and missing levels/lines could have a significant impact on the flux-distribution and radiative acceleration. Because of such missing lines (Fe II!), the gravity derived for late B and A supergiants (T_{eff} < 13,000 K) might be underestimated by 0.1 to 0.2 dex, and we advise to refrain from an analysis of such objects with the current Fe model atom.

  In v11, on the other hand, the structure is derived by means of a CMF transfer within the range between `wavblue` and `wavred` (Sect. III/1), where explicit and background lines are merged in the underlying line list. Thus, if the considered *explicit element* would include an adequate (unpacked) list of transitions (e.g., via `LINES**.dat`), the radiative transfer (and thus fluxes, g_{rad}, etc) would be correct, and thus the atmospheric structure. The same would be true (without almost any change within the current program) if the element was treated *as a background one*, and the levels and depacked transitions were updated in the atomic data base for background elements (see below).

- By calculating large grids, Hopf parameters could be created in dependence of atmospheric parameters, and automatically read into the program. Please contact Sergio Simon Diaz for progress in this respect.

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[5] In `fastwind_params`, this number (`nlev`) is currently set to 50, but could be increased without problems to 99.

[6] Here, we use *all* elements in the background approach, including those that are otherwise treated as explicit ones

- Test (again) whether fluxes and mean opacities are calculated correctly when optically thick clumping has been included **(v10 only)** . Uncomment corresponding statement in `subroutine cont` in `nlte.f90`

- Optically thick clumping **(v10 only)** needs to be tested when incoherent electron scattering is adopted. Moreover, incoherent scattering needs to be included below 3000 A, and maybe also for UV-type spectral ranges.

Regarding v11 alone …

- Include optically thick clumping (package `cmf_all.f90, formalsol_add.f90`), in the same spirit as in v10.

- At least in the formal integral, Voigt broadening for background and explicit elements needs to be included for the UV-type spectral range.

- If more elements than H/He/N were included as explicit elements, please check the convergence and the results, and potentially update the Ng-extrapolation (`cmf_all.f90`).

Part IV: Updating the atomic data base for background elements

Whilst it is fairly simply to update an explicit element (by modifying the corresponding DETAIL-input and the `LINES**.dat` file), updating the background elements is more complex, and we thus provide suitable tools, in packages `make_ATOMDAT_NEW_new` and `make_ATOMDAT_2016_new`, corresponding to the identically denoted directories in `inicalc/ATOMDAT_NEW`[7]. Both packages have the same structure[8], and the scripts/programs contained therein work similarly.

Note that the original packages are quite empty, and contain only those files that are actually necessary to set up all further files. This can be achieved with `script_nl3`, wherein a variety of files (some of them particularly long) are created. All these latter files can be deleted with `script_reset`.

All background-element data can be found in directory `fw_data`, within a multitude of files for **individual ions** with ending (=*type*)

[7]after `ATOMDAT_2016_new` has been created with *make install_data* as described in Chap. I/I and I/II
[8] differences in the actual data are explained in Chap. I/I

- `.info` — data source and data quality (cf. `_README` in `fw_data`; files only present for `ATOMDAT_2016_new`, and not for `ATOMDAT_NEW_new`)
- `.lev` — energy, statistical weights and term designation for **packed** levels
- `.pho` — photo ionization data
- `.col` — CBB data
- `.nldr` — dielectronic recomb. data
- `.nl3` — RBB data, for **depacked** transitions

Not all of these file types are present for each ion, and indeed only three types (`.lev`, `.pho`, `.nl3`) are mandatory. If a file type is not present, default approximations will be used.

Format and data of the above files are (almost) self explanatory, and, in case, can be looked up in `nlte_approx.f90`. The idea of any change/update of background data is as follows.

1. Make a safety copy of the original data bases `fw_data` and `wmbasic_data`
2. Update the `.nl3` file (wavelengths and gf-values of *unpacked* levels, indices of corresponding *packed* levels) according to your preferences.
3. Check that the indices of `.nl3` and `.lev` are consistent; in case, include new levels.
4. update `.pho`, `.col`, `.nldr`, `.info` files and check conistency with `.lev` file
5. if a new ion has been created, update the ionization energies and in case the partition functions of the uppermost ion (the one without any transitions) in `wmbasic_data/ionisen` and `wmbasic_data/partit`, respectively.
6. run `script_nl3` (if using a non-Linux OS, recompile all programs and adapt the script – see top of script), and check that no error occurs. If there is an error, correct the new data files in `fw_data` (see also next item).
7. this script

   (i) (via the program `mkatom` in `build`, based on `mkatom.f90` in `atom`) checks for basic inconsistencies (check file `fort.333` in `fw_data` if generated), combines the data of all ions into five single files (one per type, excluding .info), and creates `html` files in `fw_data/html` similar to those

that can be found online (Chap. I/I/). Particularly, it creates the (combined) files `atomcol, atomgf3, atomlev, atomnldr, atompho` and `generalinfo`[9] which **refer to the packed levels**[10], and are used within `nlte_approx.f90` to construct and solve the NLTE rate equations for the packed levels (see Sect. *level packing*). Moreover, it creates the files `nl3a, nl3g, nl3i` (all binary format) and `nl3info` (ascii) in directory `fw_data/Line-List-Bin-Sou`rce, again a combination of the individual ion.nl3 files.

(ii) runs the program `meta_2.0.eo` (in `dir_atomnl2_meta`) which creates the file `atomnl2_meta_v2.0` used to solve the *approximate* NLTE equations in `nlte_approx.f90`, as well as the .ascii version of the `nl3*` files (Chap. I/II), to be found in `dir_atomnl2_meta` under name `nl3*_all.ascii`

(iii) Finally, the program `conv_ascii_add_HHelines.eo` adds some missing H and He lines, and creates the files `nl3a_all, nl3g_all, nl3i_all` (all binary format) and `nl3info_all` (ascii) in directory `fw_data_new.`

(iv) Moreover, the script copies all additionally required files to `fw_data_new`, which subsequently

8. need to be copied to the corresponding `inicalc/ATOMDAT_NEW/ATOMDAT_*`

9. After executing step 8, we are finished with the update. Of course, careful tests with the new data set is urgently required.

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[9] This file provides info about the amount of data for the packed levels within each ion (currently ≈ 140), see `subroutine atomdat` in `nlte_approx.f90.`

[10] Thus, the (packed) gf-value of a specific transition in `atomgf3` must be the sum of all individual gf_i values in the `nl3*` files that have *the same indices* as the packed transition. The wavelength of the packed transition is chosen as the wavelength of the component with the largest gf_i value.

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